Chapter 9. Bootstrap Tests

Parametric and nonparametric bootstrap tests, duality with confidence intervals

Contents

The problem

A test needs the distribution of the statistic *under* H_0, but the bootstrap mimics the *actual* F, which may not satisfy H_0. There are two ways around this.

1. **Resample under the null.** Build an estimate $\hat{F}_0$ that satisfies H_0, and resample from it.
2. **Resample a pivot.** If $T_\theta = (\hat{\theta} - \theta)/\widehat{\mathrm{se}}$ is (approximately) pivotal, its law is the same under H_0 and under the truth. Approximate it by $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\mathrm{se}}^*$ resampled from $\hat{F}$, and compare with the observed T_{θ_0}.

In both cases the p-value is $\hat{p} = \{1 + \#(T^{*b} \text{ at least as extreme as } T_{\mathrm{obs}})\}/(B+1)$.

9.1 Parametric bootstrap tests

In a model $F(\cdot; \theta, \eta)$ with nuisance η, test $H_0 : \theta = \theta_0$ with a statistic $S(\mathbb{X}, \theta_0)$ that is large under H_1.

- *Null resampling:* compute the restricted maximum likelihood estimate $\hat{\eta}_0$ under $\theta = \theta_0$, draw samples from $F(\cdot; \theta_0, \hat{\eta}_0)$, and recompute $S(\mathbb{X}^*, \theta_0)$.
- *Pivot resampling:* draw from $F(\cdot; \hat{\theta}, \hat{\eta})$ and recompute $S(\mathbb{X}^*, \hat{\theta})$.

Proposition 9.1

For $X_i \overset{\text{i.i.d.}}{\sim} N(\mu, \sigma^2)$ and $S = \sqrt{n}(\bar{X} - \mu_0)/s$, both schemes reproduce the exact null distribution t_{n-1}.

Proof. Under null resampling the bootstrap sample is $N(\mu_0, \hat{\sigma}_0^2)$, so $\sqrt{n}(\bar{X}^* - \mu_0)/s^* \sim t_{n-1}$ whatever $\hat{\sigma}_0$ is. Under pivot resampling the sample is $N(\hat{\mu}, \hat{\sigma}^2)$ and $\sqrt{n}(\bar{X}^* - \hat{\mu})/s^* \sim t_{n-1}$. ■

Exact pivots are not always available. For Laplace(μ, b), $|X_i - \mu|/b \sim \mathrm{Exp}(1)$, so $2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$ is a pivot for μ only if b is known. With b unknown one falls back on approximate pivots (e.g. the CLT-based t-statistic), or on null resampling from Laplace$(\mu_0, \hat{b}_0)$ with $\hat{b}_0 = \frac{1}{n}\sum |X_i - \mu_0|$. With two samples and a common scale, the null model fits a common center (the pooled median) and a pooled $\hat{b}_0$.

In general the two schemes agree asymptotically. Studentized statistics give more accurate tests than raw differences (Hall and Wilson). A non-pivotal statistic, such as a ratio $\hat{\theta}/\theta_0$ whose spread depends on θ, should only be used with null resampling.

9.2 Nonparametric bootstrap tests

One sample, location. For $H_0 : \theta = \theta_0$ in $X_i = \theta + \epsilon_i$:

- null resampling: resample from the recentered data $\tilde{X}_i = X_i - \hat{\theta} + \theta_0$, which satisfy H_0;
- pivot resampling: resample from the raw data and use $T^* = (\hat{\theta}^* - \hat{\theta})/\widehat{\mathrm{se}}^*$.

For a mean and the t-statistic the two are identical: recentering shifts every bootstrap mean by the same constant. A two-sided test uses $|T^*| \geq |T_{\mathrm{obs}}|$.

Two samples, $H_0 : F = G$. Pool the samples and draw both bootstrap samples from the pooled empirical distribution. This is the with-replacement analogue of the permutation test, and the two give similar answers except in very small samples.

Two samples, equal locations with possibly different shapes. Here $F \neq G$ is allowed under $H_0 : \mu_1 = \mu_2$, so pooling is wrong and permutation tests are invalid. Use pivot resampling within each sample:

$$T^* = \frac{\bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})}{\sqrt{s_X^{*2}/m + s_Y^{*2}/n}}, \qquad T_{\text{obs}} = \frac{\bar{Y} - \bar{X}}{\sqrt{s_X^2/m + s_Y^2/n}}.$$

Equivalently, recenter each sample to the common pooled mean and resample within samples. Centering both samples at a common value and then using the *unstandardized* difference without subtracting $\bar{Y} - \bar{X}$ does not give a null distribution; it reproduces the sampling distribution of the observed difference.

9.3 Duality with bootstrap intervals

Motivation. As in classical inference, a test can be obtained by checking whether the null value falls inside a confidence interval, so every bootstrap interval yields a bootstrap test.

Proposition 9.2

The two-sided level-α pivot test of $H_0 : \theta = \theta_0$ based on $\hat{\theta} - \theta$ (resp. on $(\hat{\theta} - \theta)/\widehat{\text{se}}$) accepts exactly when θ_0 lies in the basic (resp. bootstrap-t) interval.

Proof. The test accepts iff $Q_{H^*}(\frac{\alpha}{2}) \leq \hat{\theta} - \theta_0 \leq Q_{H^*}(1-\frac{\alpha}{2})$, i.e. iff $\hat{\theta} - Q_{H^*}(1-\frac{\alpha}{2}) \leq \theta_0 \leq \hat{\theta} - Q_{H^*}(\frac{\alpha}{2})$, which is (8.1). The studentized case gives (8.2) in the same way. ■

Percentile-type intervals have no simple test counterpart, because the transformation φ behind them is unknown.

9.4 Selected exercises

Motivation. The first exercise shows what goes wrong when the bootstrap distribution is used as a null distribution without recentering. The others work out an exact Laplace pivot and the correct resampling for two samples with unequal variances.

Exercise 9.2 (A bootstrap test that never rejects)

Let $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} N(\theta, 1)$, treated as coming from an unknown distribution. For $H_0 : \theta = \theta_0$ against $H_1 : \theta > \theta_0$, take $S = \bar{X} - \theta_0$, let G^* be the bootstrap distribution of $S^* = \bar{X}^* - \theta_0$ (resampled from the raw data), and use the p-value $p = 1 - G^*(S_{\text{obs}})$. Decide whether: (a) G^* approximates the null distribution of S; (b) the test controls the type I error at $\alpha = 0.05$; (c) the power at $\theta_1 > \theta_0$ is reasonable, also as $\theta_1 \to \infty$.

Solution.

Step 1. (Where G^* is centered.) $S^* = (\bar{X}^* - \bar{X}) + S_{\text{obs}}$. By Exercise 7.5(a), $\bar{X}^* - \bar{X}$ has bootstrap mean 0 and variance $\hat{\sigma}^2/n$, and by the bootstrap CLT for the mean it is approximately $N(0, \hat{\sigma}^2/n)$. So $G^* \approx N(S_{\text{obs}}, \hat{\sigma}^2/n)$. It mimics the sampling distribution of S around its observed value, $N(\theta - \theta_0, 1/n)$, not the null distribution $N(0, 1/n)$. So (a) fails unless $S_{\text{obs}} \approx 0$.

Step 2. (The p-value does not depend on θ.) $p = P^*(\bar{X}^* > \bar{X})$ is unchanged when every x_i is shifted by a constant, so it is a function of $(x_1 - \bar{x}, \dots, x_n - \bar{x})$ only. Under $N(\theta, 1)$ the law of these residuals does not depend on θ. Hence the distribution of p is the same for every θ, and $P_{\theta_1}(p \le \alpha) = P_{\theta_0}(p \le \alpha)$ exactly, for every n and every θ_1.

Step 3. (Its size.) By Step 1, $p = P^*(\sqrt{n}(\bar{X}^* - \bar{X})/\hat{\sigma} > 0) \to 1 - \Phi(0) = \frac{1}{2}$ in probability. So $P(p \le 0.05) \to 0$.

Conclusion: (b) holds only trivially, since the test essentially never rejects. (c) fails completely: the power equals the size for every θ_1, even as $\theta_1 \to \infty$. The remedy is either to resample from data recentered at θ_0, or to use the pivot $S^* - S_{\text{obs}} = \bar{X}^* - \bar{X}$ and compare it with S_{obs} (Section 9.2). ■

Exercise 9.5 (An exact Laplace pivot)

Let $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} \text{Laplace}(\mu, b)$, with density $\frac{1}{2b} e^{-|x-\mu|/b}$, and b known.
(a) Show that $|X_i - \mu|/b \sim \text{Exp}(1)$ and $V(\mu) = 2\sum_i |X_i - \mu|/b \sim \chi^2_{2n}$.
(b) Show that $E|X - c| = d + be^{-d/b}$ with $d = |c - \mu|$, so $E|X - c|$ is uniquely minimized at $c = \mu$.
(c) Conclude that rejecting $H_0 : \mu = \mu_0$ when $V(\mu_0) > q_{1-\alpha}$, the $(1-\alpha)$-quantile of χ^2_{2n}, is an exact level-α test that is consistent against every $\mu \neq \mu_0$.

Solution [Supplement].

(a) With $Y = X - \mu$, $P(|Y|/b > t) = 2\int_{bt}^{\infty} \frac{1}{2b} e^{-y/b}\, dy = e^{-t}$ for $t \ge 0$. Then $2|X_i - \mu|/b \sim \chi^2_2$, and the sum of n independent copies is χ^2_{2n}, as in Exercise 1.7(a).

(b) By symmetry take $c = \mu + d$ with $d \ge 0$. Use $|Y - d| = (Y - d) + 2(d - Y)^+$ and $(d - Y)^+ = \int_{-\infty}^{d} I(Y \le t)\, dt$. With $P(Y \le t) = \frac{1}{2}e^{t/b}$ for $t < 0$ and $1 - \frac{1}{2}e^{-t/b}$ for $t \ge 0$,

$$E(d - Y)^+ = \int_{-\infty}^{0} \tfrac{1}{2} e^{t/b}\, dt + \int_{0}^{d} (1 - \tfrac{1}{2} e^{-t/b})\, dt = d + \tfrac{b}{2} e^{-d/b}.$$

Hence $E|Y - d| = -d + 2d + be^{-d/b} = d + be^{-d/b}$. Its derivative in d is $1 - e^{-d/b} > 0$ for $d > 0$, so it is strictly increasing in $d = |c - \mu|$.

(c) Under H_0, $V(\mu_0) \sim \chi^2_{2n}$ by (a), so $P(V(\mu_0) > q_{1-\alpha}) = \alpha$ exactly. Under $\mu \neq \mu_0$, the law of large numbers gives $V(\mu_0)/(2n) \to E|X - \mu_0|/b > 1$ a.s., by (b). On the other hand $q_{1-\alpha}/(2n) \to 1$, since $\chi^2_{2n}/(2n) \to 1$ in probability. So $P(V(\mu_0) > q_{1-\alpha}) \to 1$. With b unknown, $V(\mu_0)$ is no longer computable, which is why one turns to the resampling schemes of Section 9.1. ■

Exercise 9.6 (Two samples with unequal variances)

Let $X_1, \dots, X_m$ have variance σ_X^2 and $Y_1, \dots, Y_n$ variance σ_Y^2, independent, with $H_0 : \mu_X = \mu_Y$.
(a) Recenter both samples at a common value c: $\tilde{X}_i = X_i - \bar{X} + c$, $\tilde{Y}_j = Y_j - \bar{Y} + c$. Show that, drawing the same resampling indices, the studentized statistic computed from $(\tilde{X}^*, \tilde{Y}^*)$ equals the pivot T^* of Section 9.2 computed from (X^*, Y^*).
(b) Show that resampling both samples from the pooled data gives, in large samples, $\text{Var}^*(\bar{Y}^* - \bar{X}^*) \approx \sigma_X^2/n + \sigma_Y^2/m$, whereas $\text{Var}(\bar{Y} - \bar{X}) = \sigma_X^2/m + \sigma_Y^2/n$.

Solution [Supplement].

(a) Resampled means shift with the data: $\bar{\tilde{X}}^* = \bar{X}^* - \bar{X} + c$ and $\bar{\tilde{Y}}^* = \bar{Y}^* - \bar{Y} + c$. So $\bar{\tilde{Y}}^* - \bar{\tilde{X}}^* = \bar{Y}^* - \bar{X}^* - (\bar{Y} - \bar{X})$. Sample variances are shift invariant, so $\tilde{s}_X^{*2} = s_X^{*2}$ and $\tilde{s}_Y^{*2} = s_Y^{*2}$. Hence the two statistics coincide replicate by replicate, whatever c is.

(b)

Step 1. Let $N = m + n$. Under H_0 the pooled data have population mean $\approx \mu$, and by the law of large numbers their empirical variance is

$$\hat{\sigma}^2_{\text{pool}} \approx \frac{m\sigma_X^2 + n\sigma_Y^2}{N}.$$

Step 2. Both bootstrap samples are drawn independently from the pooled empirical distribution, so by Exercise 7.5(a),

$$\text{Var}^*(\bar{Y}^* - \bar{X}^*) = \hat{\sigma}^2_{\text{pool}}\Big(\frac{1}{m} + \frac{1}{n}\Big) \approx \frac{m\sigma_X^2 + n\sigma_Y^2}{N} \cdot \frac{N}{mn} = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}.$$

Step 3. The weights of σ_X^2 and σ_Y^2 are interchanged compared with $\text{Var}(\bar{Y} - \bar{X})$. The two agree iff $m = n$ or $\sigma_X^2 = \sigma_Y^2$. Otherwise pooled resampling, like a permutation test, has the wrong null spread: it is anticonservative when the smaller sample has the larger variance. This is why within-sample resampling as in (a) is required. ■